

A disproof of Conjecture 00000002041

A set of density $1/4$ whose subset sums miss every $n \equiv 2, 3 \pmod{4}$

1 The conjecture

Conjecture 00000002041 reads (English version):

Definition: A complete sequence satisfies $\sum_{a \in A, a \leq n} \geq n$. *Conjecture:* Every set of density ≥ 0.24 is a complete sequence covering minimally; the threshold 0.24 can be improved to an explicit $(1 - 1/e)$ type bound.

The Chinese version says the same: a complete sequence is a sequence whose partial sums satisfy $\sum_{a \in A, a \leq n} \geq n$; every set of density ≥ 0.24 is a minimal covering by a complete sequence; and the threshold 0.24 can be improved to an explicit bound of $(1 - 1/e)$ type.

Result. The set $A = \{1\} \cup \{4k : k \geq 1\}$ has density $1/4 \geq 0.24$ under each of the enumerated density readings. It fails each of the enumerated readings of “complete sequence (covering minimally)”, including the usual universal reading of the displayed definition. So the first clause is false under these readings, and with it the conjunction. Every step is a Lean 4/Mathlib theorem (Section 6). One variant is excluded and is *not* refuted: the eventual summand- a condition, which A satisfies for all $n \geq 8$ (Section 6).

2 Reading

The definition is garbled: the summand of Σ is missing, “density” is unqualified, and “covering minimally” is undefined. We therefore formalize the enumerated readings below and the usual universal reading of the displayed definition ((C7): for all $n \geq 1$), and refute all 4×8 combinations with the single witness A .

Density ≥ 0.24 (list **densityReadings**): (D1) the natural density exists and is ≥ 0.24 ; (D2) the lower density is ≥ 0.24 ; (D3) the upper density is ≥ 0.24 ; (D4) the Schnirelmann density is ≥ 0.24 .

Complete sequence (covering minimally) (list **completenessReadings**): (C1) complete: every $n \geq 1$ is a sum of distinct elements; (C2) eventually complete: every sufficiently large n is such a sum; (C3) minimal complete: complete, and no element can be removed; (C4)–(C6) contains a complete, a minimal complete, or an eventually complete subset; (C7) the stated condition with summand a : $\sum_{a \in A, a \leq n} a \geq n$ for all $n \geq 1$; (C8) the stated condition with summand 1 : $\#\{a \in A : a \leq n\} \geq n$ for all $n \geq 1$.

The conjecture is the conjunction of the first clause with the “ $(1 - 1/e)$ -improvement” clause. The main theorem refutes the first clause conjoined with an arbitrary proposition P . Section 5 also refutes every threshold $\theta \leq 1$ (in particular $1 - 1/e$) under (D1)/(D2) with (C1).

3 Definitions and sources

$A(n) = \#(A \cap \{1, \dots, n\})$ (`count`, the same expression Mathlib’s `schnirelmannDensity` uses). Natural density (`HasNatDensity`): $A(n)/n \rightarrow \delta$. Wikipedia (*Natural density*): “If this probability tends to some limit as n tends to infinity, then this limit is referred to as the asymptotic density of A .” The lower and upper densities (`lowerDensity`, `upperDensity`) are $\liminf$ and $\limsup$ of $A(n)/n$. The Schnirelmann density is Mathlib’s `schnirelmannDensity`, $\inf_{n \geq 1} A(n)/n$. Wikipedia (*Schnirelmann density*): “where $A(n)$ denotes the number of elements of A not exceeding n and $\inf$ is infimum.”

Complete sequence (`IsCompleteSeq`; Wikipedia, *Complete sequence*): “a sequence of natural numbers is called a complete sequence if every positive integer can be expressed as a sum of values in the sequence, using each value at most once.” Minimality (`IsMinimalComplete`; same article): “This sequence is minimal, since no value can be removed from it without making some natural numbers impossible to represent.” Sums of distinct elements are `IsSubsetSum` $S \ n := \exists \text{ finite } T \subseteq S \text{ with } \sum_{a \in T} a = n$.

4 The proof

Lemma 1 (`count_A`, `count_A_bounds`). *For $n \geq 1$, $A(n) = 1 + \lfloor n/4 \rfloor$, so $\frac{1}{4} \leq A(n)/n \leq \frac{1}{4} + \frac{1}{n}$.*

Proof. $A \cap [1, n] = \{1\} \sqcup \{x \in [1, n] : 4 \mid x\}$, and the second set has $\lfloor n/4 \rfloor$ elements (Mathlib `Nat.Ioc_filter_dvd_card_eq_div`). Also $4\lfloor n/4 \rfloor \leq n < 4\lfloor n/4 \rfloor + 4$. $\square$

Theorem 2 (`A_hasNatDensity`, `A_lowerDensity`, `A_upperDensity`, `A_schnirelmann`, `A_dense`). *A has natural density $1/4$, lower and upper density $1/4$, and Schnirelmann density $\geq 1/4$. Hence (D1)–(D4) hold.*

Proof. Squeeze $A(n)/n$ between $1/4$ and $1/4 + 1/n \rightarrow 1/4$. A convergent sequence has $\liminf = \limsup =$ its limit. The bound $A(n)/n \geq 1/4$ for all $n \geq 1$ is exactly `le_schnirelmannDensity_iff`. $\square$

Lemma 3 (`sum_mod_four`, `not_subsetSum`). *Every sum of distinct elements of A is $\equiv 0$ or $1 \pmod{4}$. Hence no $n \equiv 2, 3 \pmod{4}$ is such a sum.*

Proof. For finite $T \subseteq A$, every element of $T \setminus \{1\}$ is a multiple of 4, and 1 occurs at most once. So $\sum T \equiv [1 \in T] \pmod{4}$. $\square$

Theorem 4 (`A_not_complete_any`). *A fails (C1)–(C8).*

Proof. (C1): 2 is not representable. (C2): $4N + 2 \geq N$ is not representable. (C3) implies (C1). (C4)–(C6): a subset $B \subseteq A$ has fewer subset sums, so 2 and all $4N + 2$ are not representable from B (`subset_not_complete`, `subset_not_eventuallyComplete`). (C7): the elements ≤ 2 are $\{1\}$, with sum $1 < 2$. (C8): their number is $1 < 2$. $\square$

5 Supplement: no threshold works

$B = \mathbb{N} \setminus \{2\}$ has $B(n) = n - 1$ for $n \geq 2$, so its natural and lower density equal 1. But 2 is not a sum of distinct elements of B : any such summand is ≤ 2 and $\neq 2$, so it lies in $\{0, 1\}$, and the sum is at most 1. Hence for every $\theta \leq 1$, “every set of natural (or lower) density $\geq \theta$ is complete” is false (`no_threshold`). This includes $\theta = 1 - 1/e$.

6 Lean 4 formalization

File `lean/Conjecture2041/Basic.lean` (namespace `C2041`, 269 lines, Lean v4.33.1, Mathlib v4.33.1, only import `Mathlib`). Main theorem (ASCII rendering):

```
theorem conjecture_2041_false (P : Prop) :  
  forall D in densityReadings, forall C in completenessReadings,  
    not ((forall S : Set Nat, D S -> C S) /\ P)
```

Here `densityReadings` is the list (D1)–(D4) and `completenessReadings` is the list (C1)–(C8) of Section 2, as predicates on `Set ℕ`. The proof instantiates $S := A$ (`A_dense`, `A_not_complete_any`). Supplement: `no_threshold` ($\theta : \mathbb{R}$) ($h\theta : \theta \leq 1$).

Axiom audit. `lake build` succeeds, and the file also compiles with `-DwarningAsError=true`. `#print axioms for conjecture_2041_false and no_threshold` reports `[propext, Classical.choice, Quot.sound]`. There is no `sorry`, no `native_decide` and no added axiom.

Scope note (excluded variant). The *eventual* summand- a condition, “ $\sum_{a \in A, a \leq n} a \geq n$ for all sufficiently large n ”, is excluded and is not refuted: A satisfies it. For $n \in [4k, 4k+3]$ the elements of A up to n are $1, 4, \dots, 4k$, with sum $1 + 2k(k+1)$, and $1 + 2k(k+1) \geq 4k+3$ for $k \geq 2$. So the condition holds for every $n \geq 8$; it fails only at $n = 2, 3, 6, 7$. Every infinite set of positive density satisfies this variant, which would make the 0.24 threshold meaningless. The refutation covers the enumerated readings (D1)–(D4) and (C1)–(C8), which include the usual universal reading (C7) of the displayed definition.